

Md Ashikur Rahman

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RESEARCH INTERESTS

Trustworthy multimodal machine learning; vision-language models; AI safety and robustness.

EDUCATION

American International University-Bangladesh

2011 - 2015

B.Sc. in Computer Science and Engineering

- CGPA: 3.87/4.00; top 3%; *magna cum laude*
- Merit Scholarship and Tuition Fee Waiver

PEER-REVIEWED PUBLICATIONS

[P1] Abdullah Ibne Hanif Arean, Niamul Hassan Samin, Md Arifur Rahman, Renu Akter Sweety, Juena Ahmed Noshin, **Md Ashikur Rahman***. "Stroke-Level Connectivity Verification: Grounding Vision-Language Models Against Topology Hallucination in Diagram Understanding." *International Conference on Document Analysis and Recognition (ICDAR)*, 2026. Accepted for oral presentation.

*Corresponding author. [\[Code\]](#)

PREPRINTS AND MANUSCRIPTS UNDER REVIEW

[M1] **Md Ashikur Rahman**, Md Arifur Rahman, Niamul Hassan Samin, Khandaker Rifah Tasnia, Sifat Rahman Ahona, Juena Ahmed Noshin. "Beyond Aggregate Risk: Role-Stratified Conformal Risk Control for LLM Tool Calls." Manuscript under review, 2026.

[M2] **Md Ashikur Rahman**, Juena Ahmed Noshin, Niamul Hassan Samin, Abdullah Ibne Hanif Arean, Md Hasibul Amin, Md Arifur Rahman. "When Detector-Based Grounding Metrics Measure Vocabulary: A Cautionary Audit of Entity Claims in Video-QA Reasoning Traces." Manuscript under review, 2026.

[M3] **Md Ashikur Rahman**, Md Arifur Rahman, Nusrat Jahan Trisna, Juena Ahmed Noshin. "Math-Encoded Jailbreaks Across Provider-Matched Models and Inference-Time Reasoning Configurations." Manuscript under review, 2026.

ADDITIONAL PUBLICATION

[A1] "Automated Detection of Diabetic Retinopathy Using Deep Residual Learning." *International Journal of Computer Applications*, 2020.

PROFESSIONAL EXPERIENCE

Lead AI Engineer, The KOW Company

Jan 2023 - Present

Research

- Lead applied research on AI safety, multimodal reliability, vision-language grounding and hallucination, uncertainty estimation, and risk-controlled LLM tool use.
- Designed a role-stratified conformal risk-control framework for heterogeneous LLM tool calls [M1].
- Conducted a cautionary audit of detector-based grounding metrics on video question-answering reasoning traces [M2].
- Designed evaluations of math-encoded jailbreaks across provider-matched models and inference-time reasoning configurations [M3].
- Developed stroke-level connectivity verification for topology hallucination in diagram understanding; last and corresponding author on an ICDAR 2026 oral paper [P1].
- Develop evaluation frameworks, benchmarks, and reproducible pipelines for multimodal grounding, hallucination analysis, and model reliability.

Engineering Leadership

- Lead a multidisciplinary engineering team building and deploying multimodal AI products spanning image generation, virtual try-on, computer vision, and 3D reconstruction (Retouched.ai, Omnimage.ai, HoloSnap.ai, CogniX, and The Fitting Room).
- Provide technical leadership across machine-learning architecture, code review, offline evaluation, production deployment, and cross-functional delivery.

RESEARCH MENTORING

- Mentor and supervise four researchers on literature review, research planning, experimental design, reproducible implementation, result analysis, and academic writing.
- Advise junior collaborators through manuscript preparation and submission on projects in vision-language grounding, LLM safety, and risk-controlled tool use.

RESEARCH ARTIFACTS

- **LogicBench-1K**: Diagram benchmark and supporting artifacts for evaluating topology hallucination and structural grounding; released with the ICDAR 2026 paper [P1]. Role: research lead and corresponding author; contributions include problem formulation, benchmark design oversight, evaluation methodology, and manuscript leadership.

PRIOR EXPERIENCE

Senior Machine Learning Engineer, The KOW Company

Jul 2021 - Dec 2022

- Improved object-detection and segmentation performance by 20-35% on internal benchmarks; resulting models were deployed through Retouched.ai.
- Led more than six client machine-learning engagements and built offline evaluation and production A/B testing workflows for model quality and reliability.

Machine Learning Engineer, The KOW Company

Jul 2020 - Jun 2021

- Developed production deep-learning models and scalable training and evaluation pipelines for object recognition, segmentation, and background removal.

Senior Software Engineer, Smart Technologies (BD) Ltd.

May 2017 - Dec 2019

- Led development of enterprise supply-chain systems, including architecture and optimization of a 4 TB SQL Server deployment; reduced report generation from approximately 20 minutes to under one minute.

Software Engineer, Proggasoft

Mar 2015 - Aug 2016

- Developed production web applications and backend systems for DevSkill.com using ASP.NET MVC and SQL.

AWARDS AND INVITED TALKS

- Champion, BASIS National ICT Awards, 2020 (Retouched.ai)
- Finalist, Asia Pacific ICT Alliance Awards, 2021
- Artificial Intelligence in Advertising, invited workshop speaker, Daffodil International University

TECHNICAL SKILLS

Research Methods: Experimental design; benchmark and dataset design; ablation studies; error analysis; conformal prediction and risk control; calibration and uncertainty evaluation; grounding and hallucination evaluation

Multimodal AI: Vision-language models; visual grounding; multimodal reasoning; image-text alignment; LLM safety and jailbreak evaluation

Tools: Python; PyTorch; Hugging Face Transformers and Diffusers; Jupyter; Git; experiment tracking and reproducible pipelines